Phase I SBIR Cost Volume

Topic ARM26TX04-NV001

Volume 3: Cost Volume

ReluLayer — A Deterministic `#![no_std]` Rust Safety Governor

Atelier Logic Group LLC dba ReluLayer™

Unique Entity Identifier: G9WFN3AL69J3

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Cost Volume Summary

This Volume 3 Cost Volume presents the basis of estimate (BOE) and detailed cost breakdown for the Atelier Logic Group LLC Phase I effort under Army SBIR Topic ARM26TX04-NV001. The proposed price is **$200,000.00**, including fixed fee of **$18,226.40** (10.027% on total estimated cost). All labor hours are aligned with the 6-month technical work plan described in Volume 2.

1 Direct Labor Breakdown

Table 1: Phase I Direct Labor Breakdown

Role	Hours	Rate (USD)	Amount (USD)
Principal Investigator (Ishmael Ibrahim Amin)	480	95.00	$45,600.00
Senior Embedded Engineer	360	85.00	$30,600.00
Subtotal Direct Labor	**840**	—	**$76,200.00**

The 840 direct labor hours correspond to the six-month Phase I schedule (Tasks 1–4): requirements and architecture, kernel design and fixed-point ABI, host and ARM HIL validation, and equivalence testing with final reporting. The PI provides systems architecture and SBIR direction; the Senior Embedded Engineer implements the Rust kernel, C ABI, and bare-metal target integration.

2 Materials and Equipment

Table 2: Phase I Direct Materials and Equipment

Item	Qty	Unit	Unit Cost (USD)	Amount (USD)
Pixhawk 6X / STM32H743 Development Rigs	3	each	$833.33	$2,500.00
SEGGER J-Link Ultra+ / ST-Link V3 Probes	2	each	$900.00	$1,800.00
USB-Saleae High-Speed Logic Analyzer	1	each	$1,200.00	$1,200.00
Subtotal Materials and Equipment	—	—	—	**$5,500.00**

Hardware quantities are sized for the 6-month HIL validation on the ARM Cortex-M7 (STM32H743 / Pixhawk 6X). Multiple rigs enable parallel host/target equivalence and fault-injection experiments; the debug probes and logic analyzer support deterministic timing verification of the 128-cycle kernel.

3 Indirect Cost Pools and Rates

Table 3: Cost Pool Summary

Pool	Rate	Base	Amount (USD)
Direct Labor	—	840 hours (PI + Sr. Embedded Eng.)	$76,200.00
Fringe Benefits	30.00%	Direct Labor	$22,860.00
Overhead	40.00%	Direct Labor + Fringe	$39,624.00
Materials and Equipment	—	3 rigs, 2 probes, 1 analyzer	$5,500.00
Other Direct Costs (ODCs)	—	Cloud Emulation & HIL Digital Twin Test Infrastructure	$13,880.00
General and Administrative (G&A)	15.00%	Total Direct + Overhead + Materials + ODCs	$23,709.60
Total Estimated Cost Pools	—	—	**$181,773.60**

Fringe covers payroll taxes, health insurance, and retirement benefits. Overhead covers facilities, utilities, computing infrastructure, and project management tools. G&A captures corporate administration, accounting, contracts, and proposal support. Other Direct Costs (ODCs) are separately tracked for cloud emulation and HIL digital-twin test infrastructure. The rates are provisional for fiscal year 2026 and will be finalized in the cost accounting disclosure statement at award.

4 Summary Budget and Price

Table 4: Phase I Cost and Price Summary

Cost Element	Amount (USD)
Direct Labor	$76,200.00
Fringe Benefits (30.00%)	$22,860.00
Overhead (40.00%)	$39,624.00
Materials and Equipment	$5,500.00
Other Direct Costs (ODCs)	$13,880.00
Subtotal Directs + Overhead + Materials	**$158,064.00**
General and Administrative (15.00%)	$23,709.60
Total Estimated Cost	**$181,773.60**
Fixed Fee (10.027%)	$18,226.40
Total Phase I Proposal Price	**$200,000.00**

5 Basis of Estimate Narrative

5.1 Labor Hour and Rate Basis

The labor estimate is derived from the 6-month Phase I technical work plan in Volume 2. The Principal Investigator (480 hours) leads architecture, customer coordination, and the final report. The Senior Embedded Engineer (360 hours) implements the `#![no_std]` Rust kernel, C-compatible FFI, host regression harness, and ARM Cortex-M7 target build. Hourly rates reflect fiscal year 2026 loaded salaries for the National Capital Region and prevailing rates for defense SBIR safety-critical software work.

5.2 Equipment and Material Basis

Equipment costs are based on publicly available manufacturer and distributor pricing for Pixhawk 6X-class development kits, professional JTAG/SWD debug probes, and high-speed USB logic analyzers. Quantities maintain at least one active host/target equivalence testbed and one independent fault-injection testbed throughout the period, plus a spare configuration backup.

5.3 Other Direct Costs (ODCs) Basis

The $13,880.00 Other Direct Costs line funds Cloud Emulation and HIL Digital Twin Test Infrastructure required to execute the eight design-of-experiments (DOEs) in Volume 2 at realistic scale. This includes on-demand cloud compute for Monte Carlo SIL parameter sweeps, containerized firmware-in-the-loop regression runners, and persistent artifact storage for host/target equivalence traces. The estimate is sized for six months of moderate cloud consumption (compute, egress, and object storage) and will be tracked as a discrete direct cost line item. Without this infrastructure, deterministic 1,000-iteration HIL equivalence and swarm-scale DOE-05/DOE-07 stress tests would be impractical within the Phase I period.

5.4 Indirect Rate and Fixed Fee Basis

Fringe is applied to direct labor; overhead is applied to direct labor plus fringe; G&A is applied to the subtotal of direct labor, fringe, overhead, materials, and ODCs. The rates are provisional for fiscal year 2026 and will be documented in the certified cost or pricing data. The fixed fee is 10.027% of total estimated cost, equal to $18,226.40, within the SBIR Phase I fee ceiling.

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